



Tanuj Chauhan

2026PHZ8140

JRF (PhD)

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Profile

Early-stage doctoral researcher focusing on solar magnetohydrodynamics (MHD), with an M.Sc. in Physics from IIT Guwahati. Possesses a strong background in programming, mathematical physics, and theoretical physics. Passionate about astrophysics and cosmology, and highly motivated to apply computational methods to complex problems in solar physics.

Objective

To advance my early doctoral research in solar magnetohydrodynamics (MHD) and leverage my background in theoretical physics to make impactful contributions to the broader field of astrophysics.

Education[#]

Year	Degree / Certificate	Board /Institute	Major	CGPA/Percentage%
<i>Current</i>	<i>PhD</i>	<i>IIT Delhi</i>	<i>Astrophysics</i>	-
2025	Masters Of Science	IIT Guwahati	Physics	7.78
2023	Bachelors Of Science	Himachal Pradesh University	Physics	8.91
2019	Senior secondary	CBSE	Physical Science	88.2
2017	Secondary	CBSE	GENERAL	9.6

Academic Achievements:

- **JEST Physics (2026)**
Qualified GATE Physics exam with AIR 83.
- **GATE Physics (2026)**
Qualified GATE Physics exam with AIR 94.
- **JEST Physics (2025)**
Qualified GATE Physics exam with AIR 351.
- **IIT JAM Physics (2023)**
Qualified IIT Jam Physics exam with AIR 258.
- **B.Sc. III, 1st in Mathematics (2023)**

Projects & Dissertation

- **Study of Magnetohydrodynamics in Accretion Flow around Black Holes** - Investigated the global structure of advection-dominated accretion flows around Schwarzschild black holes using a magnetohydrodynamic (MHD) framework. Computed global transonic solutions to analyse the formation of centrifugally supported shock waves, mass-loss effects, and post-shock corona properties under the influence of radiative cooling and toroidal magnetic fields.

(Under Prof. Santabrata Das, Indian Institute of Technology, Guwahati)

- **Path Integral formulation of Quantum Mechanics** Analysed quantum systems using path integral and operator approaches to link kernels to physical observables and calculate many-body partition functions.
(Under **Prof. Justin R. David, Indian Institute of Science, Bangalore**)
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Workshops & Seminars:

- **Frontier Areas in Material Research (2022)**
Participated in a workshop based on Frontier Areas in Material Research 2022 conducted by the Indian Academy of Sciences.
 - **International Workshop Srinivasa Ramanujan: the man beyond infinity (2020)**
Attended online International Workshop organised by Central University of Himachal Pradesh.
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Key courses taken:

- **Physics:** Electrodynamics, Modern Physics, Nuclear & Particle Physics, Thermal Physics & Stats. Quantum Mechanics - (I,II) , Special Theory of Relativity
 - **Maths:** Linear Algebra, Differential Calculus, Differential Equations, Numerical Methods
 - **Computer Science:** Problem solving using Python, Data Structures & Processing, Database Management System.
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Position/POR:

- Core Team member at the Equinox Astronomy Club of IIT Guwahati.
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Interest areas:

- Cosmology
 - Astrophysics
 - Relativity
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Hobbies and interests:

- Astrophotography
 - Writing and discussions
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Technical skills

- Programming languages: Python, FORTRAN
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Personal details:

- Date of birth: 17 JAN 2002
- Sex: Male
- Nationality: Indian
- Languages known: English and Hindi



Declaration

- I hereby declare that above mentioned information is true to my knowledge and belief.
- I bear the responsibility for the correctness of above-mentioned particulars.